Ricardo Arcesio Villanueva Montenegro

Full-Stack Engineer · Applied AI and multi-agent systems · Distributed, high-volume systems

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Summary

Full-Stack Engineer with 3 years of experience building production applications and distributed systems with TypeScript, React, Next.js, Vue, NestJS and Node.js on Google Cloud and AWS. Designed a personalized nutrition and training plan engine with a multi-agent architecture (Vertex AI, OpenAI, Mastra) and event-driven processing over Pub/Sub for a B2B2C platform deployed across four major Spanish retail chains, within a project awarded a 300,000 EUR European grant. Built and shipped to production 10 data acquisition integrations processing over 1.5 million documents, with 20+ hour runs on AWS Fargate without failures. Created the agent-orchestrated development platform I use daily: 38 skills, 11 agents and automated gates that reproduce CI before every commit. Experienced with NestJS microservices using Clean Architecture, microfrontends (single-spa, Vue 3), asynchronous messaging (RabbitMQ, Pub/Sub), CI/CD and 9 months of product ownership as a Product Owner.

Experience

Magnar **Aug 2026 – Present**

Scraper Reliability Engineer (Oct 2026 to present) · Scraper Developer (Aug to Oct 2026) *Chile (Remote)*

- **Built and shipped 10 data acquisition integrations to production** across public portals in three countries, with a combined corpus of **over 1.5 million documents** and 20+ hour runs on AWS Fargate without failures.
- **Reverse engineering of opaque systems**: undocumented APIs, undeclared pagination caps, filters the backend accepts and silently ignores. On one source I diagnosed that blocking was by IP rather than by request type, and that its result set truncated without warning; neither was documented and both reshaped the architecture.
- **Data identity and idempotency design**: immutable uniqueness keys chosen by measuring coverage and collisions against the real corpus. A poorly chosen key silently drops records and cannot be corrected afterwards.
- **Production reliability**: triage and diagnosis from logs and failure signatures, separating code issues from infrastructure and source issues. I detect *false greens*, processes that finish «successfully» without producing anything, and close a case only when the next scheduled run comes back green with data, not when the PR is merged.
- **Cost-aware decisions**: HTTP-first architecture, with headless browser and proxy as a last resort rather than a default, since traffic is the bulk of the bill.
- *Stack:* TypeScript, Node.js, Puppeteer/Playwright, Cheerio, pdf-lib, Tesseract, PostgreSQL, AWS Fargate and S3, Docker, GitHub Actions.

Southteams *(client: Qurable · SKY Plus, SKY Airline's loyalty program)* **Sep 2026 – Dec 2026**

Software Developer *Remote*

- **Full-stack development on a distributed production platform** (microfrontends, web applications and microservices) for the accounts and loyalty ecosystem: registration, authentication, profile, points and benefits.
- **Organization-scale frontend**: microfrontends with single-spa and Vue 3, plus an Astro web monolith with content managed from a headless CMS, in multiple languages and per market.
- **NestJS microservices with Clean Architecture** and asynchronous messaging, maintaining contracts between services and with external partners.
- **End-to-end incident diagnosis**: I reproduce, trace the flow across services and fix at the layer where the problem originates, not where it surfaces.
- I take part in technical and functional refinement with product, UX and business, turning requirements into concrete scope and proposing architectural alternatives, and verify in QA and Stage against acceptance criteria, with evidence, before promoting to production.
- Contributed to the team's **internal agent tooling**: the shared context and memory layer across agents and iterations, with idempotent seeding, conflict detection and read-back verification.
- *Stack:* TypeScript, Vue 3, Astro, Tailwind, single-spa, NestJS, RabbitMQ, MongoDB, GitLab CI, Jira.

GUNDO Health and Food S.L. **Aug 2024 – Jul 2026**

Product Owner & Full-Stack Engineer (Nov 2025 to Jul 2026) · Full-Stack Engineer (Aug 2024 to Nov 2025) *Spain (Remote)*

- Worked on the Fitness area of an AI-powered white-label B2B2C health and nutrition platform, deployed as an own-brand product for Uvesco, Ametller Origen, DIA and CrossFit.
- Developed **GENIE**, personalized nutrition based on genomic, microbiome and biochemical data, **winner of the European DRG4FOOD program (300,000 EUR grant)** in a consortium with ADN Institut (Spain) and i3S (Portugal).
- Designed the training and nutrition plan generation engine with a **multi-agent architecture** (Mastra, Vertex AI, OpenAI) and asynchronous event-driven processing over Google Cloud Pub/Sub.
- Built a **computer vision meal scanner**: ingredient recognition and macronutrient calculation from images, with recommendations and alerts based on the user's profile.
- Evolved the platform to a **multi-tenant architecture** and hardened API security (CORS, JWT, App Check, cross-user access prevention). Designed adaptive questionnaires with branching logic plus admin and IAM modules.
- **As Product Owner (nine months)**: defined the technical and functional roadmap, managed sprints and releases, prioritized with metrics and A/B testing, and owned CI/CD on Google Cloud and AWS.

Gelvez Distribuciones S.A.S. **Nov 2023 – Jul 2024**
Full-Stack Developer, IT Assistant *Cúcuta, Colombia*

- Automated administrative processes that consumed hours of manual work: supplier report delivery (Python, Selenium/Playwright) and bulk validation of business partners against public databases, for regulatory compliance.
- Developed a **company intranet** (Python/Django) that reduced ERP dependency and operating costs, plus a REST API for investment requests (Flask + Firebase) and a monitoring dashboard (Vite + React).

Systems I built

Agent orchestration platform for software development. I built the system I work with daily: a layer on top of Claude Code that drives the research, build, review and delivery cycle, with **38 skills and 11 specialized agents**.

- **Mechanical gates that reproduce CI locally before committing**, with a hook that blocks the operation if the gate is not green. An auditor agent reproduces the judgment portion by reading the repository's actual CI prompt, so when the team changes its rules, the gate changes with them.
- **Autonomous loop with gates and re-entry**: it advances on its own and stops only on scope questions, round limits, or findings that change the problem.
- **Versioned doctrine** extracted from real operation, state persisted across sessions, and a dedicated variant for production operations with read-only monitor access over API and credential redaction.

Projects

- **Shelvora** (shelvora.rickdev.tech): **offline-first** Android reading app (React Native New Architecture + Hermes, Expo, TypeScript). **Content-addressed storage** so the library does not break when a source changes or disappears, **hybrid translation** online with an on-device fallback model and a glossary to keep proper nouns consistent across chapters, and a multi-source engine covering 7 sites. Built with an **AI-orchestrated plan-first workflow**: 16 phases, one branch and one PR per phase.
- **Rival**: gym app with a competitive ranked ladder. ELO derived from the Wilks score and a seasonal tier system. Main contributor (∼94% of activity), including the PR that introduced the **AI-first spec-driven architecture** the whole codebase runs on. Expo, Hono, Drizzle, PostgreSQL, BetterAuth, RevenueCat, Turborepo.
- **InkLens**: AR tattoo preview, with 3D designs rendered live on skin via on-device body tracking (MediaPipe), without uploading the photo to a server unless the user opts in. Next.js 14, Three.js/React Three Fiber, Turborepo, Stripe, Cloudflare R2.
- **Atelierum AI**: turns a photo of a garment into a cut-ready sewing pattern (SVG/DXF/PDF) using computer vision to estimate measurements; reduces a 2-4 hour manual task to ∼20 seconds. Next.js + FastAPI, 130+ tests.
- **Others**: DineFlow (restaurant management, Express + Firebase), HandleMinder (personal finance, React + FastAPI), 3D Portfolio (rickdev.tech, React + Three.js), chunk-learner (language learning with AI-assisted voice practice), e-commerce REST API in Go.

Skills

- **Languages:** TypeScript, JavaScript, Python, SQL, Go
- **Frontend:** React, Next.js, Vue 3, Astro, React Native (New Architecture, Hermes), Expo, Tailwind, Three.js, Vite, microfrontends (single-spa)
- **Backend:** Node.js, NestJS, Express, Hono, FastAPI, Django, Flask, REST APIs, microservices, Clean Architecture, asynchronous messaging (RabbitMQ, Pub/Sub)
- **Databases:** PostgreSQL, MongoDB, MySQL, Firestore, BigQuery, SQLite, Drizzle ORM
- **Cloud and DevOps:** AWS (Fargate, ECR, S3, CDK), Google Cloud (Cloud Build, Pub/Sub, Firestore, BigQuery), Firebase, Docker, Kubernetes, CI/CD (GitHub Actions, GitLab CI), Git, Turborepo, Sentry, PostHog
- **AI and agents:** multi-agent system design, agent-orchestrated development, Claude Code (plugins, skills, hooks, subagents), Vertex AI, OpenAI, Gemini Vision, Mastra, ML Kit, MediaPipe
- **Data at scale:** API reverse engineering, scraping and automation (Puppeteer, Playwright, Cheerio, Selenium), anti-blocking mitigation (WAF, captcha, IP rotation), PDF extraction and OCR, deduplication and record identity
- **Product and process:** refinement with business and UX, roadmap, sprints and releases, acceptance criteria and QA evidence, technical documentation
- **Payments and Auth:** Stripe, MercadoPago, RevenueCat, BetterAuth, JWT

Education and certifications

- **Systems Engineering**, UNAD, Colombia (Jan 2025 to Dec 2027, in progress) · **Software Analysis and Development Technologist**, SENA, Colombia (Apr 2022 to Jul 2024)
- **Anthropic:** AI Fluency Framework & Foundations (2026) · Claude Code in Action (2026) · Claude Code 101 (2026). **Cisco:** JavaScript Essentials 1 (2024) · Python Essentials 1 and 2 (2022-2023)